Department of Computer Science
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SUBJECT NAME:	**COMPUTER ORGANIZATION AND ARCHITECTURE**
SUBJECT CODE:	**SECR 1033**
SEMESTER:	**2 – 2023/2024**
LAB TITLE:	**Lab 2: Arithmetic Equations & Operations**
STUDENT INFO :	Execute the lab in group of two. **Student 1**: *No. 1, 3, 5* **Student 2**: *No 2, 4, 6* **Name 1:** **Chau Ying Jia** **Metric No:** **A23CS0213** **Name 2:** **Tan Zhi Ming** **Metric No:** **A23CS0189**
SUBMISSION DATE:	_____

MARKS:

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Arithmetic Equation Coding in Assembly Language

Q1. Execute the program below. Determine output of the program by inspecting the content of the related registers.
a) Fill in Table 1 with the content of each register or variable on every LINE, in **Hexadecimal** (as per the output). Please complete the comments for every LINE.
b) Paste the screenshot of all registers' content after each LINE is executed.

```
INCLUDE Irvine32.inc
.data
var1 word 1
var2 word 9

.code
main PROC
        mov ax, var1     ; LINE1
        mov bx, var2     ; LINE2
        xchg ax, bx      ; LINE3
        mov var1, ax     ; LINE4
        mov var2, bx     ; LINE5
        call DumpRegs
        exit
main ENDP
END main
```

Answer Q1

a) Fill (Write) in the contents for the related register in each line:

Table 1

LINE1	**AX = 0001h** **var1 = 0001h**	Move the value of var1 (1d) into register AX
LINE2	**BX =** **var2 =**	
LINE3	**AX =** **BX =**	
LINE4	**AX =** **var1 =**	
LINE5	**BX =** **var2 =**	

b) Paste here screenshot of all registers' content after each LINE is executed:

LINE1:

```
EAX = 010F0001 EBX = 00F86000 ECX = 00D010AA EDX = 00D010AA ESI = 00D010AA
  EDI = 00D010AA EIP = 00D03666 ESP = 010FFE48 EBP = 010FFE54 EFL = 00000246
```

LINE2:

```
EAX = 00AF0001 EBX = 00900009 ECX = 00D010AA EDX = 00D010AA ESI = 00D010AA
  EDI = 00D010AA EIP = 00D0366D ESP = 00AFF818 EBP = 00AFF824 EFL = 00000246
```

LINE3:

```
EAX = 00AF0009 EBX = 00900001 ECX = 00D010AA EDX = 00D010AA ESI = 00D010AA
  EDI = 00D010AA EIP = 00D0366F ESP = 00AFF818 EBP = 00AFF824 EFL = 00000246
```

LINE4:

```
EAX = 00AF0009 EBX = 00900001 ECX = 00D010AA EDX = 00D010AA ESI = 00D010AA
  EDI = 00D010AA EIP = 00D03675 ESP = 00AFF818 EBP = 00AFF824 EFL = 00000246
```

Name	Value	Type
ax	0x0009	unsigned ..
bx	0x0001	unsigned ..
var1	0x0009	unsigned ..
var2	0x0009	unsigned ..

LINE5:

```
EAX = 00AF0009 EBX = 00900001 ECX = 00D010AA EDX = 00D010AA ESI = 00D010AA
  EDI = 00D010AA EIP = 00D0367C ESP = 00AFF818 EBP = 00AFF824 EFL = 00000246
```

Name	Value	Type
ax	0x0009	unsigned ...
bx	0x0001	unsigned ...
var1	0x0009	unsigned ...
var2	0x0001	unsigned ...

Q2. Execute the program below. Determine output of the program by inspecting the content of the related registers and watches.
a) Fill in Table 2 with the content of each register or variable on every LINE, in **Hexadecimal** (as per the output). Please complete the comments for every LINE.
b) Paste the screenshot of all registers' content after each LINE is executed.

Arithmetic expression: Rval = (-Xval + (Yval – Zval)) + 1

```
include irvine32.inc

.data
Rval DWORD ?
Xval DWORD 26
Yval DWORD 30
Zval DWORD 40

.code
main proc
      mov eax,Xval       ; LINE1
      neg eax            ; LINE2
      mov ebx,Yval       ; LINE3
      sub ebx,Zval       ; LINE4
      add eax,ebx        ; LINE5
      inc eax            ; LINE6
      mov Rval,eax       ; LINE7
      exit
main endp
end main
```

Answer Q2

a) Fill (Write) in the contents for the related register in each line:

Table 2

LINE1	**EAX = 0000001Ah** **Xval = 0000001Ah**	Move the value of Xval (26d) into register EAX
LINE2	**EAX =**	
LINE3	**EBX =** **Yval =**	
LINE4	**EBX =** **Zval =**	
LINE5	**EAX =** **EBX =**	
LINE6	**EAX =**	
LINE7	**EAX =** **Rval =**	

b) Paste here screenshot of all registers' content after each LINE is executed:

LINE1:

```
Registers
EAX = 0000001A EBX = 00E69000 ECX = 00151005 EDX = 00151005 ESI = 00151005 EDI = 00151005 EIP = 00151015 ESP = 00DDFB0C EBP = 00DDFB18 EFL = 00000246
```

LINE2:

```
Registers
EAX = FFFFFFE6 EBX = 00E69000 ECX = 00151005 EDX = 00151005 ESI = 00151005 EDI = 00151005 EIP = 00151017 ESP = 00DDFB0C EBP = 00DDFB18 EFL = 00000293

0x00154008 = 0000001E
```

LINE3:

```
Registers
EAX = FFFFFFE6 EBX = 0000001E ECX = 00151005 EDX = 00151005 ESI = 00151005 EDI = 00151005 EIP = 0015101D ESP = 00DDFB0C EBP = 00DDFB18 EFL = 00000293

0x0015400C = 00000028
```

LINE4:

```
Registers
EAX = FFFFFFE6 EBX = FFFFFFF6 ECX = 00151005 EDX = 00151005 ESI = 00151005 EDI = 00151005 EIP = 00151023 ESP = 00DDFB0C EBP = 00DDFB18 EFL = 00000287
```

LINE5:

```
Registers
EAX = FFFFFFDC EBX = FFFFFFF6 ECX = 00151005 EDX = 00151005 ESI = 00151005 EDI = 00151005 EIP = 00151025 ESP = 00DDFB0C EBP = 00DDFB18 EFL = 00000283
```

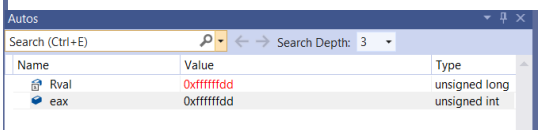
LINE6:

```
Registers
EAX = FFFFFFDD EBX = FFFFFFF6 ECX = 00151005 EDX = 00151005 ESI = 00151005 EDI = 00151005 EIP = 00151026 ESP = 00DDFB0C EBP = 00DDFB18 EFL = 00000287

0x00154000 = 00000000
```

LINE7:

```
Registers
EAX = FFFFFFDD EBX = FFFFFFF6 ECX = 00151005 EDX = 00151005 ESI = 00151005 EDI = 00151005 EIP = 0015102B ESP = 00DDFB0C EBP = 00DDFB18 EFL = 00000287
```



Q3. Execute the program below. Determine output of the program by inspecting the content of the related registers.
a) Fill in Table 3 with the content of each register or variable on every LINE, in **Hexadecimal** (as per the output). Please complete the comments for every LINE.
b) Paste the screenshot of all registers' content after each LINE is executed.

Arithmetic expression: var4 = [(var1 * var2) + var3] - 1

```
include irvine32.inc

.data
var1 DWORD 5
var2 DWORD 10
var3 DWORD 20
var4 DWORD ?

.code
main proc
      mov eax, var1          ; LINE1
      mul var2               ; LINE2
      add eax, var3          ; LINE3
      dec eax                ; LINE4
      exit
main endp
end main
```

Answer Q3

a) Fill (Write) in the contents for the related register in each line:

Table 3

LINE1	**EAX = 00000005h** **var1 = 00000005h**	Move the value of var1 (5d) into register EAX
LINE2	**EAX =** **var2 =**	
LINE3	**EAX =** **var3 =**	
LINE4	**EAX =** **var4 =**	

b) Paste here screenshot of all registers' content after each LINE is executed:

LINE1:

```
EAX = 00000005 EBX = 00C38000 ECX = 00AB1005 EDX = 00AB1005 ESI = 00AB1005
  EDI = 00AB1005 EIP = 00AB1015 ESP = 00BFF7E4 EBP = 00BFF7F0 EFL = 00000246

0x00AB4004 = 0000000A
```

LINE2:

```
EAX = 00000032 EBX = 00C38000 ECX = 00AB1005 EDX = 00000000 ESI = 00AB1005
  EDI = 00AB1005 EIP = 00AB101B ESP = 00BFF7E4 EBP = 00BFF7F0 EFL = 00000202

0x00AB4008 = 00000014
```

LINE3:

```
EAX = 00000046 EBX = 00C38000 ECX = 00AB1005 EDX = 00000000 ESI = 00AB1005
  EDI = 00AB1005 EIP = 00AB1021 ESP = 00BFF7E4 EBP = 00BFF7F0 EFL = 00000202
```

LINE4:

```
EAX = 00000045 EBX = 00C38000 ECX = 00AB1005 EDX = 00000000 ESI = 00AB1005
  EDI = 00AB1005 EIP = 00AB1022 ESP = 00BFF7E4 EBP = 00BFF7F0 EFL = 00000202
```

Q4. Execute the program below. Determine output of the program by inspecting the content of the related registers.
a) Fill in Table 4 with the content of each register or variable on every LINE, in Hexadecimal (as per the output). Please complete the comments for every LINE.
b) Paste the screenshot of all registers' content after each LINE is executed.

Arithmetic expression: var4 = (var1 * 5) / (var2 – 3)

```
include irvine32.inc
.data
var1 WORD 40
var2 WORD 10
var4 WORD ?
.code
main proc
mov ax,var1             ; LINE1
mov bx,5          ; LINE2
mul bx            ; LINE3
mov bx,var2             ; LINE4
sub bx,3          ; LINE5
div bx            ; LINE6
mov var4,ax             ; LINE7
exit
main endp
end main
```

Answer Q4

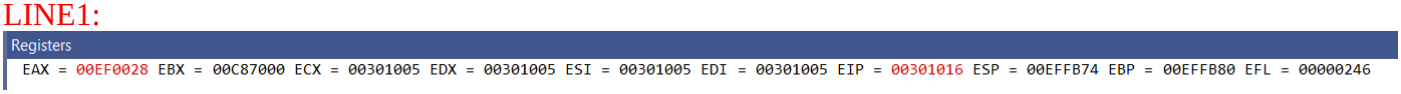
a) Fill (Write) in the contents for the related register in each line:

Table 4

LINE1	**AX = 0028h** **var1 = 0028h**	Move the value of var1 (40d) into register AX
LINE2	**BX =**	
LINE3	**AX =** **BX =**	
LINE4	**BX =** **var2 =**	
LINE5	**BX =**	
LINE6	**AX =** **BX =** **DX =**	
LINE7	**AX =** **var4 =**	

b) Paste here screenshot of all registers' content after each LINE is executed:

LINE1:



LINE2:

Registers

```
EAX = 00EF0028 EBX = 00C80005 ECX = 00301005 EDX = 00301005 ESI = 00301005 EDI = 00301005 EIP = 0030101A ESP = 00EFFB74 EBP = 00EFFB80 EFL = 00000246
```

LINE3:

Registers

```
EAX = 00EF00C8 EBX = 00C80005 ECX = 00301005 EDX = 00300000 ESI = 00301005 EDI = 00301005 EIP = 0030101D ESP = 00EFFB74 EBP = 00EFFB80 EFL = 00000202
```

LINE4:

Registers

```
EAX = 00EF00C8 EBX = 00C8000A ECX = 00301005 EDX = 00300000 ESI = 00301005 EDI = 00301005 EIP = 00301024 ESP = 00EFFB74 EBP = 00EFFB80 EFL = 00000202
```

LINE5:

Registers

```
EAX = 00EF00C8 EBX = 00C80007 ECX = 00301005 EDX = 00300000 ESI = 00301005 EDI = 00301005 EIP = 00301028 ESP = 00EFFB74 EBP = 00EFFB80 EFL = 00000202
```

LINE6:

Registers

```
EAX = 00EF001C EBX = 00C80007 ECX = 00301005 EDX = 00300004 ESI = 00301005 EDI = 00301005 EIP = 0030102B ESP = 00EFFB74 EBP = 00EFFB80 EFL = 00000202
```

LINE7:

Registers

```
EAX = 00EF001C EBX = 00C80007 ECX = 00301005 EDX = 00300004 ESI = 00301005 EDI = 00301005 EIP = 00301031 ESP = 00EFFB74 EBP = 00EFFB80 EFL = 00000202
```

Name	Value	Type
ax	0x001c	unsigned short
var4	0x001c	unsigned short

Short Notes for MUL CX and DIV BL:

MUL CX

a. MUL always uses AX (or its extended versions EAX or RAX) as the implicit destination register.
b. The operand size determines the size of the result:
    i. Byte-sized operand: Result in AX
    ii. Word-sized operand: Result in DX:AX
    iii. Doubleword-sized operand (32-bit mode): Result in EDX:EAX
    iv. Quadword-sized operand (64-bit mode): Result in RDX:RAX
c. The upper half of the result (DX or EDX or RDX) holds any overflow bits.
d. The Carry Flag (CF) is set if the upper half of the product is non-zero.

DIV BL

a. DIV always uses the DX:AX or EDX:EAX pair as the implicit dividend register.
b. The divisor is specified as the operand of the DIV instruction.
c. The quotient is stored in AX (for 16-bit division) or EAX (for 32-bit division).
d. The remainder is stored in DX.
e. Clear DX (or EDX for 32-bit division) before division to ensure a correct 16-bit or 32-bit dividend.
f. If the divisor is 0, a division error occurs.
g. The Overflow Flag (OF) is set if the quotient is too large to fit in the destination register.

Q5. Given the following instructions as is Code Snippet 1.

a) Write a full program to execute the Code Snippet 1.
b) What are the contents of the related registers after Code Snippet 1 is executed? Paste the screenshot of DumpReg.

; Code Snippet 1 (MUL CX)

```
MOV DX,0        ; Clear DX
MOV AX,1000h    ; Load 1000h into AX
MOV CX, 25h     ; Load 25h into CX
MUL CX          ; Multiply AX by CX, storing the result in DX:AX
```

Answer Q5

a) Screenshot of full program (.asm) :

lab2_ques5 (Global Scope)

```
1   TITLE lab2
2   ; Author: Chau Ying Jia
3   ; Date: 19/5/2024
4   include Irvine32.inc
5
6   .code
7   main proc
8   MOV DX, 0           ; Clear DX
9   MOV AX, 1000h       ; Load 1000h into AX
10  MOV CX, 25h         ; Load 25h into CX
11  MUL CX              ; Multiply AX by CX, storing the result in DX : AX
12
13  call DumpRegs
14  exit
15  main endp
16  end main
```

b) Paste here the screenshot of the final registers' content (DumpReg):

```
EAX = 006F5000 EBX = 0053F000 ECX = 00F60025 EDX = 00F60002 ESI = 00F61005
  EDI = 00F61005 EIP = 00F6101F ESP = 006FF7B0 EBP = 006FF7BC EFL = 00000A07
```

Q6. Given the following instructions as is Code Snippet 2.

a) Write a full program to execute the Code Snippet 2.
b) What are the contents of the related registers after Code Snippet 2 is executed? Paste the screenshot of DumpReg.

```
; Code Snippet 2 (DIV BL)
MOV DX, 0       ; Clear DX to form the 16-bit dividend in DX:AX
MOV AX, 803h    ; Load the dividend (8003h) into AX
MOV BL, 10h     ; Load the divisor (10h) into BL
DIV BL          ; Divide DX:AX by BL, whereby AX=quotient & DX=remainder
```

Answer Q6

a) Screenshot of full program (.asm) :

labwork2 (Global Scope)

```
1   TITLE lab2
2   ; Author: TAN ZHI MING
3   ; Date: 19 / 5 / 2024
4   include Irvine32.inc
5
6   .code
7   main proc
8   MOV DX, 0               ; Clear DX to form the 16 - bit dividend in DX : AX
9   MOV AX, 803h            ; Load the dividend(8003h) into AX
10  MOV BL, 10h             ; Load the divisor(10h) into BL
11  DIV BL                  ; Divide DX:AX by BL, whereby AX=quotient & DX=remainder
12  call dumpRegs
13  exit
14  main ENDP
15  END main
16
```

b) Paste here the screenshot of the final registers' content (DumpReg):

```
EAX=01160380  EBX=012A3010  ECX=010010AA  EDX=01000000
ESI=010010AA  EDI=010010AA  EBP=0116FA48  ESP=0116FA3C
EIP=01003671  EFL=00000246  CF=0  SF=0  ZF=1  OF=0  AF=0  PF=1
```

Registers

```
EAX = 00190380 EBX = 002B8010 ECX = 009110AA EDX = 00910000 ESI = 009110AA EDI = 009110AA EIP = 00913671 ESP = 0019FBF4 EBP = 0019FC00 EFL = 00000246
```